

Second-Hand Product Escrow Marketplace |

Problem Statement #39

Use Case ID	UC-06	Use Case Name	Approve Delivery (Sign-off)
Primary Actor	Buyer		
Secondary Actors	Inspection Agent, Payment Gateway, Admin (in alternate flow)		
Trigger	Buyer opens a delivered order and chooses to confirm or dispute receipt of the product.		
Preconditions	1) Order exists and is in 'Delivered' state. 2) Payment is locked in escrow (UC-04). 3) Inspection Agent has filed a sign-off report (UC-05). 4) Buyer is registered and logged in (UC-01).		
Postconditions	Success: Order state = 'Completed'; escrow funds released to Seller (UC-08). Alternate: Order state = 'Disputed'; escrow funds frozen; dispute ticket opened for Admin (UC-07, UC-09).		

Main Success Scenario

1. Buyer opens the order details page after the product has been marked delivered.
2. System displays the Inspection Agent's sign-off report alongside the delivered item and the original listing.
3. Buyer compares the physical product against the inspection report and the listing description.
4. Buyer selects "Approve Delivery."
5. System checks that the order is in the 'Delivered' state and that escrow funds are still locked.
6. System transitions the order state from 'Delivered' to 'Approved.'
7. System invokes UC-08 (Release Escrow Funds) — an «include» relationship — instructing the Payment Gateway to transfer the held funds to the Seller.
8. Payment Gateway processes the transfer and returns a confirmation to the system.
9. System sets the order state to 'Completed' and sends a confirmation notification to both Buyer and Seller.
10. Use case ends in success.

Alternate Flow — A1: Buyer Raises a Dispute Instead of Approving

- A1.1. Instead of approving, the Buyer selects "Raise Dispute" — UC-07, an «extend» of this use case.
- A1.2. System immediately freezes the order's escrow funds and blocks any withdrawal or release attempt.
- A1.3. System sets the order state to 'Disputed' and opens a dispute ticket, notifying the Admin.
- A1.4. Admin investigates (UC-09): reviews the inspection report, order history, and any evidence submitted by the Buyer and Seller.
- A1.5. Admin decides the outcome — funds are released to the Seller, or refunded to the Buyer.
- A1.6. System updates the order to its final state ('Completed' or 'Refunded') and notifies both parties.
- A1.7. Use case ends. Control does not return to the main success scenario.

Exception Flow — E1: Payment Gateway Failure

- E1.1. At step 8, the Payment Gateway fails to process the transfer (timeout or rejection).
- E1.2. System keeps the order in 'Approved' state with funds still locked in escrow, and logs the failure.
- E1.3. System retries the transfer automatically; if it fails after 3 attempts, the order is flagged for Admin review.

Business Rules

- Escrow funds can only move to 'Released' from the 'Approved' state, never directly from 'Delivered.'
- A dispute, once raised, always takes precedence over a pending approval — funds are frozen the moment a dispute is filed.
- Only the Buyer on the order may approve delivery or raise a dispute for that order.